Memory Architectures and Long-Context Mechanisms for Long-Horizon Large Language Model Agents: A Comprehensive Survey

1 Introduction

The emergence of Large Language Model (LLM) agents capable of executing long-horizon tasks necessitates a paradigm shift from transient context processing to robust, persistent memory architectures. Long-horizon agency is formally characterized by tasks requiring sustained interaction, delayed reward acquisition, and multi-stage planning that fundamentally exceed the capacity of standard fixed-length context windows. While recent architectural innovations have successfully extended context windows to millions of tokens [1; 2], empirical evidence suggests that mere sequence length expansion is insufficient for genuine long-term understanding. Studies indicate that models often struggle to utilize information distributed across vast contexts, exhibiting a "lost-in-the-middle" phenomenon where relevance decays for tokens situated between the prompt boundaries [3; 4]. Furthermore, benchmarks such as BABILong and RULER reveal that current models effectively utilize only a fraction of their available context when faced with complex reasoning dependencies, often failing to retrieve disparate facts scattered across millions of tokens [5; 6].

This survey distinguishes critically between native long-context mechanisms and external memory systems, positing that optimal agent performance requires a synergistic integration of both. Native approaches, including efficient attention patterns [7; 8] and advanced positional encodings [9; 10], offer seamless integration but suffer from quadratic computational scaling or attention dilution. Conversely, external memory systems decouple storage from computation, enabling infinite horizon tracking through retrieval-augmented generation and vector databases. However, naive retrieval often introduces noise and disrupts temporal coherence, necessitating sophisticated management policies for writing, consolidation, and forgetting [11; 12]. The dichotomy between parametric memory, encoded within model weights, and non-parametric memory, stored in dynamic buffers, further complicates the landscape, requiring hybrid semi-parametric architectures to balance static world knowledge with dynamic episodic experience [13].

The scope of this survey encompasses a comprehensive taxonomy of these mechanisms, progressing from foundational cognitive inspirations to cutting-edge architectural innovations. We analyze how biological concepts of working, episodic, and semantic memory map onto computational modules, informing the design of hierarchical agents capable of subgoal tracking and mental time-travel [14; 15]. Special attention is devoted to dynamic management strategies that address the stability-plasticity dilemma, ensuring agents can adapt to non-stationary environments without catastrophic forgetting [16; 17]. Moreover, we evaluate the efficacy of these mechanisms in facilitating high-level planning and self-reflection, noting that while LLMs often fail at autonomous planning, memory-augmented frameworks significantly enhance heuristic guidance and error correction [18; 19]. By synthesizing insights from efficient attention, retrieval augmentation, and continual learning, this work aims to establish a unified theoretical framework for developing agents that possess not just extended context, but true long-term cognitive continuity.

2 Conceptual Foundations and Taxonomy of Agent Memory

2.1 Cognitive Architectures and Biological Inspirations for Agent Memory

The theoretical grounding for agent memory in long-horizon Large Language Model (LLM) systems is increasingly derived from computational cognitive science, specifically by mapping biological memory systems—working, episodic, semantic, and procedural—to distinct architectural modules.

This bio-inspired approach addresses the inherent limitations of standard transformer attention, which struggles with the "context window bottleneck" when processing sequences extending beyond immediate interaction windows. By emulating the hierarchical and multi-timescale dynamics of the human brain, researchers have developed agents capable of sustaining coherence, planning, and adaptation over indefinite horizons.

A primary focus of this translation is the distinction between transient working memory and persistent long-term storage. In biological systems, Baddeley's model of working memory posits a central executive coordinating an episodic buffer; computationally, this maps to mechanisms where LLMs maintain active subgoals and intermediate reasoning states while offloading detailed histories to external buffers. The [20] framework exemplifies this by leveraging subgoals as memory chunks, hierarchically managing working memory to reduce redundancy and improve success rates in long-horizon tasks. Similarly, [12] proposes a modular cognitive architecture where structured action spaces interact with internal memory components, mirroring the separation of control and storage found in symbolic AI and human cognition. This separation is critical, as unstructured linear histories often lead to performance degradation, whereas structured working memory allows for efficient context switching and task decomposition.

Beyond working memory, the integration of episodic and semantic memory draws heavily from hippocampal-neocortical interactions. The hippocampal indexing theory suggests that the hippocampus rapidly encodes specific episodes while the neocortex slowly consolidates semantic knowledge. [21] operationalizes this by synergizing LLMs with knowledge graphs and Personalized PageRank to mimic this dual-process, enabling efficient multi-hop reasoning that outperforms standard retrieval-augmented generation. Furthermore, [22] demonstrates that constructing dynamic memory graphs integrating semantic and episodic traces allows agents to navigate complex interactive environments more effectively than those relying on unstructured history. The emergence of "induction heads" in transformers, which behaviorally mirror the contextual maintenance and retrieval (CMR) models of human episodic memory, provides a mechanistic bridge between biological theory and neural network internals [23].

Procedural memory, responsible for skill acquisition and automated action sequences, is reflected in architectures that separate fast-adapting policies from slow-consolidating knowledge. The [24] framework treats LLMs as semi-parametric reinforcement learning agents, utilizing an experience memory to update policies without retraining parametric weights, thus embodying the stability-plasticity dilemma resolution seen in biological learning. Additionally, bio-inspired learning rules, such as Hebbian context gating and exponentially decaying task signals, have been shown to mitigate catastrophic forgetting in continual learning scenarios, allowing agents to retain old skills while acquiring new ones [25]. The [26] algorithm further explores this by using optimization at inference time to generate dynamic task representations, inspired by thalamocortical circuits.

Despite these advances, challenges remain in perfectly aligning artificial architectures with biological fidelity. While models like [27] incorporate attention-gated memory tagging to handle multi-timescale dependencies, they often struggle with hierarchical tasks requiring simultaneous retention of high-level stimuli and low-level details. Future directions must focus on refining these mappings, particularly in developing unified frameworks that seamlessly integrate the predictive coding principles of hierarchical temporal memory [28] with the generative capacities of modern LLMs. Ultimately, the convergence of cognitive architectures and deep learning promises to yield agents that not only process long contexts but truly understand and reason over temporal dynamics akin to human cognition.

2.2 The Parametric and Non-Parametric Memory Dichotomy

Building upon the theoretical blueprints of cognitive architectures outlined previously, the practical realization of long-horizon agency hinges on the architectural dichotomy between parametric and non-parametric memory. This distinction separates knowledge implicitly encoded within fixed model weights from information explicitly stored in dynamic, addressable buffers, forming the foundational tension in agent design. Parametric memory, realized through the static parameters of pre-trained large language models (LLMs), offers high-density storage of world knowledge and procedural skills but suffers from inherent rigidity; once trained, this "frozen" memory cannot adapt to novel contexts or correct factual errors without computationally prohibitive retraining or fine-tuning [29]. This limitation is particularly acute in long-horizon scenarios where agents must integrate time-sensitive information, leading to issues of temporal degradation and hallucination when relying solely on implicit weight-based retention [30].

Conversely, non-parametric memory mechanisms decouple storage from computation, utilizing external databases, vector stores, or internal hidden states to enable infinite context expansion and real-time updatability. Approaches such as Retrieval-Augmented Generation (RAG) and explicit read-write modules allow agents to inject specific historical trajectories or factual triplets into the context window dynamically, bypassing the capacity constraints of the transformer's attention mechanism [31; 32]. Recent innovations have evolved beyond simple retrieval to include structured symbolic memories, such as SQL databases or knowledge graphs, which support complex multi-hop reasoning and maintain belief consistency over extended interactions [33; 34]. Furthermore, latent space memory pools, as seen in [35], introduce a fixed-size, self-updatable memory component within the model architecture itself, bridging the gap by allowing weight-adjacent updates without full backpropagation.

However, non-parametric approaches introduce distinct trade-offs regarding retrieval fidelity and computational latency. While they offer limitless capacity, their efficacy is contingent upon the precision of retrieval algorithms, which often struggle with the "lost-in-the-middle" phenomenon or fail to retrieve semantically relevant but lexically distant information [36]. To mitigate these deficiencies, hybrid semi-parametric architectures have emerged as a dominant paradigm, synergistically fusing the generalized reasoning priors of parametric weights with the specific, updatable logs of non-parametric stores. For instance, [37] proposes a distributed episodic memory that enables one-shot knowledge updates, while [21] mimics hippocampal indexing to integrate new experiences into a graph structure without catastrophic forgetting. Similarly, [38] organizes tokens into episodic events using Bayesian surprise, enabling efficient retrieval over practically infinite contexts.

The future of long-horizon agency lies not in choosing between these paradigms but in optimizing their integration. Emerging trends point towards "configurable foundation models" where functional modules are dynamically assembled based on task demands, effectively treating parametric knowledge as reusable bricks and non-parametric stores as flexible working memory [39]. Ultimately, achieving robust long-term agency requires mechanisms that can seamlessly consolidate fleeting non-parametric experiences into stable parametric representations, mirroring the biological process of systems consolidation to ensure both adaptability and stability [40; 41]. This synergistic integration sets the stage for the subsequent analysis of how these storage paradigms are operationalized across varying temporal scales, from immediate token-level attention to indefinite historical consolidation.

2.3 Temporal Scales and Granularity of Information Retention

The temporal dimension of agent memory necessitates a hierarchical taxonomy ranging from immediate token-level attention to indefinite historical consolidation, addressing the fundamental trade-off between information fidelity and retention duration. At the shortest scale, standard transformer architectures exhibit a “sharp nearby, fuzzy far away” phenomenon, where models maintain high sensitivity to word order within recent contexts (approximately 50 tokens) but degrade distant history into a rough semantic field [36]. This intrinsic decay implies that native attention mechanisms function effectively as short-term working memory but fail to capture long-range dependencies without architectural intervention. While recent advances in positional interpolation and extrapolation, such as those enabling context windows up to 32k tokens, mitigate hard length limits [9; 42], empirical evaluations indicate that models still struggle to utilize information located in the middle of extended sequences, a challenge termed the “lost-in-the-middle” problem [43; 44].

Bridging the gap between transient context and permanent storage requires mid-term mechanisms capable of tracking subgoals and active states over episodes. Approaches like Expire-Span introduce learnable forgetting policies that dynamically retain critical information while expiring irrelevant states, allowing transformers to scale to tens of thousands of timesteps by optimizing the granularity of retained features [45]. Similarly, hierarchical memory architectures, such as Hierarchical Chunk Attention Memory (HCAM), enable agents to perform “mental time travel” by first attending to coarse-grained summaries of past chunks before retrieving fine-grained details, thereby decoupling retention duration from computational cost [46]. These systems mimic biological event segmentation, organizing continuous streams into discrete episodic units based on Bayesian surprise or graph-theoretic boundaries [38].

For long-term retention extending to lifelong horizons, information must undergo compression and consolidation into abstracted semantic representations or external knowledge graphs. Unlike parametric memory frozen during pre-training, non-parametric external stores allow for dynamic updates and infinite capacity [47]. Mechanisms inspired by the Ebbinghaus forgetting curve, such as those implemented in MemoryBank, modulate retention strength based on recency and significance, facilitating the transition from volatile episodic traces to stable semantic knowledge [48]. Furthermore, neurobiologically inspired frameworks like HippoRAG leverage hippocampal indexing theories to integrate new experiences into knowledge graphs efficiently, supporting multi-hop reasoning over vast historical data without catastrophic forgetting [21]. Theoretical analyses suggest that optimal memory systems for scale-free signals must sacrifice temporal precision to represent exponentially long timescales, favoring fuzzy, compressed representations over rigid shift registers [49]. Future research must therefore focus on unified architectures that seamlessly integrate these varying temporal scales, ensuring that agents can fluidly shift between high-fidelity immediate reasoning and abstracted long-term planning without incurring prohibitive computational overhead or information loss.

2.4 Taxonomy of Memory Operations and Management Policies

Building on the temporal hierarchies and storage paradigms established previously, this subsection delineates a functional taxonomy of the algorithmic primitives that govern the lifecycle of information within long-horizon agents: encoding (write), access (retrieve), maintenance (update), and eviction (forget). These operations form a closed-loop control system essential for navigating the trade-off between computational boundedness and the necessity for indefinite context retention, effectively operationalizing the theoretical distinctions between short-term attention and long-term consolidation.

The **write operation** transforms raw sensory streams and action trajectories into storable representations, addressing the fidelity-capacity trade-off inherent in the temporal scales discussed earlier. Approaches range from literal storage of episodic traces to sophisticated compression via summarization or latent projection. For instance, [50] proposes compressing long contexts into compact memory slots using an autoencoding objective, effectively distilling semantic essence while reducing token overhead. Similarly, [38] employs Bayesian surprise and graph-theoretic boundary refinement to segment continuous token streams into coherent episodic events before storage, mimicking human event cognition. In embodied settings, [51] utilizes a spatially structured 2D memory image with adaptable write operators to store arbitrary environmental information, surpassing fixed-window temporal convolutions. The critical challenge here lies in balancing fidelity with capacity; overly aggressive compression risks losing granular details necessary for future reasoning, whereas literal storage rapidly exhausts finite resources.

Complementing the write phase, **retrieval strategies** dictate how stored memories are injected back into the active context window to inform decision-making, directly mitigating the "lost-in-the-middle" phenomenon and attention decay identified in prior sections. This spectrum spans from dense vector similarity search and sparse keyword matching to learned, query-aware attention mechanisms. [52] introduces a dynamic KV cache selection algorithm that estimates token criticality based on query vectors, loading only top-k relevant pages to accelerate self-attention. Complementing this, [53] trains attention mechanisms to utilize landmark tokens for block selection, enabling random access over arbitrarily long contexts without separate retrieval modules. More complex reasoning demands iterative navigation; [54] treats the LLM as an agent that iteratively traverses a tree of summary nodes to locate relevant information, enhancing explainability and precision in long-text QA. However, retrieval efficacy often necessitates re-ranking or positional bias mitigation strategies to ensure models can effectively utilize information positioned centrally within extended sequences.

The **update and forgetting policies** constitute the regulatory mechanisms that prevent memory bloat and catastrophic interference, ensuring the system remains adaptable over lifelong horizons. Effective management requires distinguishing between transient noise and salient knowledge, a process akin to the biological consolidation and expiration mechanisms previously outlined. [45] presents Expire-Span, a learnable mechanism that assigns expiration values to hidden states, allowing the model to retain critical information while discarding irrelevant history over tens of thousands of timesteps. This aligns with the "use it or lose it" principle observed in [55], where statistical recall biases during perpetual stochastic gradient descent drive selective forgetting of rarely accessed memories. Furthermore, [56] highlights the necessity of selective elimination for invalidated information in multi-session dialogues to maintain conversational coherence. Advanced systems also incorporate consolidation, merging short-term episodic traces into stable long-term semantic structures, as seen in hierarchical frameworks that separate working and long-term memory tiers [32].

Synthesizing these operations reveals a paradigm shift from static, capacity-limited buffers to dynamic, content-addressable systems governed by adaptive policies. Future research must address the integration of these operations into unified theoretical frameworks that optimize the joint objective of retrieval precision, update efficiency, and forgetting selectivity. Emerging trends suggest a convergence toward neurobiologically inspired architectures where write, read, and forget operations are not distinct modules but emergent properties of a holistic memory dynamics system, capable of scaling to infinite horizons while maintaining temporal consistency.

3 Architectural Innovations for Native Long-Context Processing

3.1 Efficient Attention Mechanisms and Sparse Computation Patterns

The standard self-attention mechanism, while foundational to Transformer architectures, incurs a quadratic computational complexity $O(N^2)$ with respect to sequence length N, creating a prohibitive bottleneck for native long-context processing in long-horizon agents. To mitigate this, recent architectural innovations have shifted towards efficient attention mechanisms that exploit inherent sparsity patterns and dynamic selection strategies to achieve linear or sub-quadratic scaling. A primary avenue of research involves identifying and leveraging structured sparsity within attention matrices during the pre-filling stage. [7] demonstrates that long-context attention heads exhibit distinct structural patterns—specifically A-shape, Vertical-Slash, and Block-Sparse configurations—which can be detected offline and utilized to construct dynamic sparse indices. By implementing optimized GPU kernels for these specific patterns, inference latency can be reduced by up to 10x without compromising accuracy, effectively decoupling context length from computational cost.

Complementing static pattern identification, query-aware dynamic sparsity offers a more adaptive approach to token selection. [52] posits that token criticality is not intrinsic but highly dependent on the current query vector. By maintaining bounds on Key values within KV cache pages and estimating their relevance to the query, Quest selectively loads only the top-K critical pages for attention computation. This strategy yields significant speedups in self-attention operations while preserving the model's ability to reason over long dependencies. Similarly, [57] introduces a dynamic token pruning mechanism that operates across both pre-filling and decoding stages. Unlike static pruning, LazyLLM allows the model to re-evaluate and select different subsets of context tokens at each generation step, ensuring that essential information discarded in earlier steps can be recuperated if subsequently deemed relevant, thereby accelerating inference without permanent information loss.

Beyond sparsity, the integration of compressive memory mechanisms enables theoretically infinite context horizons with bounded memory footprints. [2] proposes the Infini-attention module, which augments standard masked local attention with a long-term linear attention mechanism and a compressive memory block. This hybrid design allows the model to retain a summarized history of distant tokens while focusing computational resources on local contexts, facilitating fast streaming inference for sequences extending to millions of tokens. Furthermore, [53] addresses the trade-off between compression and random access by introducing landmark tokens that represent blocks of input. This approach trains the attention mechanism to retrieve relevant blocks directly through these landmarks, maintaining the flexibility of full attention while significantly reducing the number of tokens processed per step.

Despite these advances, challenges remain in balancing the fidelity of long-range dependency modeling with the aggressiveness of sparsity. While methods like [8] utilize Locality Sensitive Hashing (LSH) to achieve near-linear time complexity even with unbounded attention entries, they introduce approximate sampling that may obscure subtle semantic nuances critical for complex agent planning. Future research must therefore focus on developing adaptive sparsity policies that dynamically adjust compression rates based on task complexity and uncertainty, ensuring that efficient attention mechanisms do not merely extend context windows but enhance the qualitative reasoning capabilities of long-horizon agents.

3.2 Advanced Positional Encoding and Length Extrapolation Strategies

While efficient attention mechanisms and architectural re-designs address the computational bottlenecks of long sequences, a foundational prerequisite for long-horizon agents remains the ability to extend the native context window of pretrained Large Language Models (LLMs) without sacrificing positional coherence. The core difficulty lies in the extrapolation of positional encodings, where models encounter index values significantly larger than those observed during pretraining, often leading to catastrophic performance degradation. A dominant paradigm for addressing this involves modifying Rotary Positional Embeddings (RoPE) through interpolation and scaling strategies. Position Interpolation (PI) [9] proposes linearly down-scaling input position indices to match the original context range, theoretically demonstrating that the upper bound of interpolation error is orders of magnitude smaller than direct extrapolation. This approach allows models like LLaMA to extend to 32k tokens with minimal fine-tuning. Building on this, YaRN [58] introduces a more nuanced scaling method that combines linear interpolation with a temperature adjustment in the attention mechanism, effectively mitigating the "sharp nearby, fuzzy far away" phenomenon and preserving short-context capabilities while enabling robust long-range dependency modeling.

Further advancements have sought to optimize the efficiency and granularity of these extensions, moving beyond simple scaling to targeted adaptation. LongLoRA [59] integrates shifted sparse attention with adapted RoPE scaling, demonstrating that context extension can be achieved with significantly reduced computational costs by focusing fine-tuning on specific attention patterns rather than dense global recalibration. Similarly, Entropy-ABF [60] identifies attention entropy stability as a critical metric, proposing a method that adjusts RoPE base frequencies and scales attention logits to adapt to larger windows with as few as 100 training samples. For extreme extrapolation scenarios, LongRoPE [1] employs an efficient search strategy to identify non-uniformities in positional interpolation, utilizing a progressive extension strategy that achieves context lengths up to 2048k tokens while recovering short-context performance through targeted readjustment.

Beyond uniform scaling, multi-scale and adaptive representations have emerged to address the "lost-in-the-middle" phenomenon, where models struggle to retrieve information located centrally within vast contexts—a challenge that persists even when computational limits are overcome. Multi-scale Positional Encoding (Ms-PoE) [3] assigns distinct scaling ratios to different attention heads, creating a fusion of short-to-long distance contexts that alleviates long-term decay effects without requiring additional fine-tuning. Complementing this, Mixture of In-Context Experts (MoICE) [61] treats different RoPE angles as experts, dynamically routing attention heads to specific contextual positions to ensure comprehensive context awareness. Alternative training-free approaches like Dual Chunk Attention (DCA) [62] decompose attention into intra-chunk and inter-chunk modules to capture relative positional information across distinct segments, achieving over 100k token support without continual training.

Despite these successes, challenges remain in balancing extrapolation capacity with the preservation of fine-grained local reasoning. While methods like Positional Skip-wisE (PoSE) [10] simulate long inputs via chunk skipping to decouple training length from target length, the theoretical limits of continuous length extrapolation via differential dynamics suggest a need for architectures that model positional scaling as a continuous function rather than discrete steps. Furthermore, while positional scaling extends the window, it does not inherently solve the quadratic complexity of the underlying attention mechanism, necessitating the sparse and compressive strategies discussed subsequently. Future research must therefore focus on unified frameworks that seamlessly integrate adaptive positional dynamics with efficient attention mechanisms, ensuring that agents can navigate infinite horizons without sacrificing the precision required for complex, multi-step reasoning tasks.

3.3 Hybrid Recurrent-Transformer Architectures and State Space Models

The quest for native long-context processing has catalyzed a conceptual convergence between recurrent neural networks (RNNs) and Transformer architectures, driven by the need to overcome the quadratic computational bottleneck of self-attention while preserving parallel training efficiency. This subsection examines the resurgence of recurrent paradigms through modern linearization techniques and the emergence of State Space Models (SSMs) as robust alternatives for capturing long-range dependencies with constant inference costs. Historically, RNNs were limited by vanishing gradients and sequential computation constraints; however, recent innovations have demonstrated that careful architectural redesign can recover the impressive performance of deep SSMs while matching their training speed [63]. By linearizing and diagonalizing the recurrence relation, models such as the Linear Recurrent Unit (LRU) achieve computational efficiency comparable to attention-based mechanisms without sacrificing the ability to model complex temporal dynamics.

A pivotal advancement in this domain is the development of Structured State Space Models, which frame sequence modeling as a continuous dynamical system discretized for digital computation. The HiPPO framework formalizes the online compression of continuous signals via projection onto polynomial bases, yielding memory update mechanisms like HiPPO-LegS that theoretically avoid priors on timescales and maintain bounded gradients [64]. Building on this, architectures such as Mamba and its variants leverage data-dependent state transitions to perform selective attention, effectively filtering irrelevant information in linear time [65]. These models address the "context dilution" problem inherent in standard linear RNNs by dynamically modulating the state space based on input content, thereby enabling precise retrieval over millions of tokens [5].

Hybrid architectures further bridge the gap by integrating recurrent memory modules directly into Transformer backbones. The Recurrent Memory Transformer (RMT) introduces segment-level recurrence, passing compressed hidden states across chunks to extend context limits while maintaining hierarchical memory structures [66]. Similarly, the xLSTM architecture revitalizes the classic LSTM by introducing exponential gating and matrix memory structures (mLSTM), rendering the mechanism fully parallelizable and competitive with state-of-the-art Transformers in both performance and scaling [67]. These hybrid designs often outperform pure attention models in tasks requiring extreme sequence lengths, as evidenced by their success on benchmarks like BABILong, where recurrent augmentations enable processing of up to 11 million tokens, far exceeding the practical limits of standard Transformers [68].

Despite these advances, trade-offs remain regarding the expressivity of linear recurrences versus the flexible associativity of full attention. While SSMs offer superior throughput and memory efficiency for streaming applications, they may struggle with certain forms of content-based retrieval that do not align with their underlying state transition dynamics. Consequently, emerging trends point toward unified frameworks that dynamically switch between attention and recurrence or employ hybrid blocks, such as those seen in multimodal settings where Mamba and Transformer layers are interleaved to balance efficiency and representational power [69]. Future research must address the theoretical bounds of these hybrid systems, particularly in optimizing the interplay between local attention windows and global recurrent states to achieve truly unlimited context horizons with minimal latency.

3.4 Chunk-Based Processing and Encoder-Decoder Context Expansion

Complementing the architectural backbone reforms discussed previously, chunk-based processing and encoder-decoder context expansion offer a distinct paradigm for extending the context horizon of decoder-only Large Language Models (LLMs). While the preceding section focused on rearchitecting the sequence modeling core to achieve linear or constant inference costs, this approach preserves the original model weights and avoids the instability of direct positional extrapolation by decomposing elongated input sequences into manageable segments. These frameworks process segments via auxiliary encoders or parallel modules to generate compressed representations, which are subsequently accessed by the frozen decoder through cross-attention mechanisms. This decoupling strategy enables theoretically infinite context windows through efficient memory management without incurring the prohibitive quadratic costs of dense self-attention on raw tokens.

A seminal approach in this domain is Context Expansion with Parallel Encoding (CEPE), which introduces a small, trainable encoder to process long document chunks independently [4]. By compressing these chunks into embedding vectors, CEPE enables a frozen decoder to leverage extended contexts via cross-attention, achieving a 10x throughput increase and reducing memory usage to one-sixth of standard baselines while extending LLaMA-2's context to 128K tokens [4]. This method effectively mitigates the "lost-in-the-middle" phenomenon by allowing the decoder to attend selectively to high-density semantic representations rather than raw token streams. Similarly, the BGE Landmark Embedding framework explores extensible embeddings where long contexts are represented by compact input units of higher information density, acting as an enhancement of typical token embedding to access a vast scope of context even with a small context window [70].

Beyond static encoding, dynamic chunk selection mechanisms have emerged to optimize context utilization by aligning with the query-aware sparsity principles hinted at in hybrid architectures. Training-free frameworks such as LongLLMLingua decompose inputs and compress prompts towards improving LLMs' perception of key information, thereby enabling reasoning over sequences far exceeding native window limits with reduced cost [71]. This aligns with the principles of query-aware sparsity, where critical tokens are identified based on their interaction with current queries, significantly reducing the computational load during the pre-filling stage [52]. Furthermore, the LongHeads framework distributes context chunks across different attention heads, allowing each head to process in-distribution length segments while collectively covering the entire long sequence, achieving linear time complexity without additional training [72].

The integration of recurrent memory concepts with chunked processing further bridges the gap between the strategies discussed in the previous section and the compression techniques outlined here. The Infini-attention mechanism incorporates compressive memory into standard transformer blocks, building both masked local attention and long-term linear attention to process streaming inputs with bounded memory [2]. This approach complements chunk-based strategies by providing a mechanism to aggregate information across chunks without retaining all historical keys and values. Additionally, hierarchical merging strategies like HOMER employ a divide-and-conquer algorithm, merging adjacent chunks at progressive transformer layers with token reduction techniques to ensure memory usage scales logarithmically with input length [73].

Despite their efficacy, these approaches face challenges regarding the fidelity of compression and the potential loss of fine-grained syntactic details during the encoding phase, a trade-off that necessitates the unified theoretical lens proposed in the following section. While methods like ICAE demonstrate that contexts can be compressed into short memory slots with minimal parameter overhead, the balance between compression rate and reasoning accuracy remains a critical area for investigation [50]. Future research directions should focus on adaptive chunking strategies that dynamically adjust segment boundaries based on semantic coherence rather than fixed token counts, and the development of bidirectional cross-attention mechanisms that allow decoders to iteratively refine their queries to encoders. Such advancements will enhance the precision of long-range dependency modeling in agent-based scenarios, setting the stage for a broader discussion on the spectrum of memory management operations governing write, store, read, and inject phases.

3.5 Unified Memory-Augmented Frameworks and Theoretical Perspectives

The proliferation of architectural innovations for long-context processing necessitates a unified theoretical lens that reinterprets diverse mechanisms—from sparse attention to recurrent state spaces—as specific instantiations of memory management operations. Rather than viewing these approaches as disjointed engineering solutions, we posit that they collectively form a spectrum of strategies governing the write, store, read, and inject phases of information flow within neural architectures. This unified perspective reveals that the fundamental challenge in long-horizon agency is not merely extending sequence length, but optimizing the trade-off between memory fidelity, computational complexity, and retrieval precision.

A primary axis of this theoretical framework distinguishes between explicit memory augmentation and implicit state compression. Approaches like [66] and [74] explicitly decouple memory storage from the primary processing stream, utilizing recurrent connections or feedback loops to maintain a bounded latent state that summarizes infinite history. These architectures effectively implement a "compressive memory" mechanism, akin to the Infini-attention proposed in [2], where local attention handles immediate context while a linear attention mechanism aggregates global statistics into a fixed-size memory block. In contrast, methods such as [53] and [73] operate by dynamically restructuring the attention landscape, treating specific tokens or merged representations as "landmarks" that facilitate random access to compressed chunks. This suggests a convergence where architectural modifications are essentially learning to identify and exploit sparsity patterns in the temporal domain, aligning with the "sharp nearby, fuzzy far away" phenomenon observed in biological and artificial systems [36].

Furthermore, the dichotomy between parametric and non-parametric memory is increasingly blurred in native long-context frameworks. While early models relied strictly on fixed weights, modern architectures like [35] and [37] introduce self-updatable memory pools within the latent space, enabling one-shot knowledge injection without full retraining. This evolution mirrors the theoretical shift towards "neural stored-program" concepts, where the model learns to manipulate its own memory weights as data [75]. Theoretical analysis indicates that such hybrid semi-parametric systems offer superior scalability for lifelong learning, as they decouple the stability of pre-trained priors from the plasticity required for episodic updates. However, this introduces complex challenges in maintaining temporal consistency; as noted in [34], unregulated memory updates can lead to contradictory belief states, necessitating rigorous reconciliation mechanisms akin to weighted MaxSAT solvers or Bayesian filtering [76].

Emerging trends point toward hierarchical and adaptive memory policies as the next frontier. Frameworks like [46] demonstrate that agents can achieve superior sample efficiency by employing coarse-to-fine retrieval strategies, attending first to high-level summaries before drilling down into episodic details. This "mental time travel" capability is theoretically grounded in the need to manage the "impossible triangle" of reliability, generalization, and locality in memory editing [29]. Future research must therefore focus on developing unified objective functions that jointly optimize for compression rates, retrieval latency, and semantic preservation. By formalizing long-context processing as a dynamic memory allocation problem, we can derive principled bounds on context capacity and guide the design of architectures that rival human-like episodic recall [38], ultimately enabling agents to operate coherently over indefinite temporal horizons.

4 External Memory Systems and Retrieval-Augmented Mechanisms

4.1 Structural Paradigms of External Memory Stores

The architectural decoupling of memory from parametric weights represents a fundamental paradigm shift in enabling long-horizon agency, moving beyond the static constraints of pre-trained knowledge to dynamic, scalable retention. This subsection categorizes external memory stores into three distinct structural paradigms: unstructured vector repositories, structured symbolic bases, and hybrid neuro-symbolic architectures, each offering unique trade-offs between retrieval fluidity and reasoning precision.

Vector-based semantic repositories constitute the most prevalent paradigm, leveraging dense embedding spaces to facilitate high-dimensional similarity search. These systems excel at associative recall, allowing agents to retrieve conceptually related historical interactions even in the absence of explicit keyword overlap. For instance, [77] demonstrates how light-weight cache-like networks can store recent hidden representations to dynamically adapt neural machine translation models to document-level contexts. Similarly, [78] utilizes latent vector spaces to encode and retrieve conceptually similar robot action episodes, enabling unsupervised prediction of future frames. However, purely vector-based approaches often lack explicit relational structure, struggling with multi-hop reasoning tasks where logical consistency is paramount. The limitation of such unstructured stores is evident in complex planning scenarios where agents fail to maintain coherent state trajectories without explicit structural constraints [79].

In contrast, symbolic and graph-structured knowledge bases prioritize logical rigor and explicit relational encoding. By storing information as entities and edges within knowledge graphs or as records in SQL databases, these systems support precise query execution and complex deduction. [33] instantiates this approach by empowering LLMs to generate SQL instructions to manipulate symbolic databases, thereby overcoming the approximate nature of neural memory to simulate complex reasoning. Furthermore, [34] introduces a symbolic memory of beliefs that utilizes a weighted MaxSAT solver to revise inconsistent beliefs, ensuring a systematic notion of belief stability. While highly accurate for structured queries, symbolic systems often suffer from rigid retrieval mechanisms that lack the semantic flexibility to handle ambiguous or noisy natural language inputs, potentially limiting their applicability in open-ended environments.

Emerging hybrid neuro-symbolic architectures seek to synthesize the fluid retrieval of vector stores with the logical constraints of symbolic graphs. These frameworks dynamically switch between associative recall and structured deduction based on task demands. [21] exemplifies this synergy by orchestrating LLMs, knowledge graphs, and Personalized PageRank algorithms to mimic the complementary roles of the hippocampus and neocortex, achieving superior performance in multi-hop question answering. Similarly, [80] combines a non-parametric graph of location connectivity with a parametric deep network for node retrieval, enabling robust navigation in unseen environments. Additionally, [30] advocates for structured explicit memory modules that improve interpretability and factual grounding compared to purely parametric or unstructured retrieval methods. Future research directions must address the integration challenges of these paradigms, particularly in developing unified interfaces that allow seamless transitions between semantic similarity and logical inference without incurring prohibitive computational overhead.

4.2 Advanced Retrieval Strategies and Context Optimization

Building upon the foundational storage substrates established in the preceding section, the efficacy of long-horizon agents hinges not merely on memory capacity but on the precision of retrieval mechanisms that govern the selection and injection of information into the active context window. As context lengths expand, naive retrieval strategies often succumb to noise accumulation and positional biases, necessitating advanced algorithmic interventions that integrate temporal, spatial, and semantic metadata. A primary challenge in this domain is the "lost-in-the-middle" phenomenon, where models struggle to attend to relevant information situated deep within extended sequences. To mitigate this, recent approaches have moved beyond simple similarity search toward order-preserving and position-aware retrieval paradigms. For instance, [81] introduces Order-Preserve Retrieval-Augmented Generation (OP-RAG), which strategically maintains the original chronological order of retrieved chunks during injection. This method counters the disruption of narrative flow and causal dependencies, demonstrating that answer quality follows an inverted U-shaped curve relative to the number of retrieved tokens, thereby identifying optimal "sweet spots" for context density that outperform brute-force long-context processing.

Furthermore, the integration of temporal and spatial metadata has emerged as a critical differentiator for agents operating in dynamic environments, addressing the limitations of static vector stores discussed earlier. Standard dense vector retrieval often overlooks the recency or physical proximity of events, leading to the retrieval of semantically similar but temporally irrelevant memories. Advanced frameworks address this by incorporating time-decay functions and event boundaries into the retrieval scoring mechanism. [38] proposes a two-stage memory process that combines similarity-based retrieval with temporally contiguous access, leveraging Bayesian surprise to segment token streams into coherent episodic events. This biologically inspired approach ensures that retrieved memories are not only conceptually related but also contextually appropriate regarding their temporal distance from the current state. Similarly, in multimodal settings, [82] highlights the necessity of prioritizing textual attention over image features during prompt prefill, suggesting that retrieval strategies must be modality-aware to efficiently compress Key-Value caches without losing critical spatial relationships.

Noise reduction and query-awareness represent another frontier in context optimization, refining how stored data is filtered before injection. The assumption that all retrieved tokens contribute equally to reasoning is increasingly challenged by evidence showing that token criticality is highly query-dependent. [52] advances this by proposing a KV cache selection algorithm that estimates the criticality of memory pages based on specific query vectors, loading only the top-k relevant pages to accelerate self-attention. This selective injection prevents the dilution of the context window with redundant data. Complementing this, [83] employs self-information metrics to filter out low-entropy, less informative content prior to injection, effectively increasing the signal-to-noise ratio within the limited context budget. Additionally, iterative retrieval methods such as [84] revise chain-of-thought steps dynamically by retrieving information relevant to both the task query and previous reasoning states, significantly reducing hallucination in complex generation tasks.

Looking forward, the convergence of these strategies points toward adaptive, multi-agent collaborative retrieval systems. As noted in [85], distributed retrieval frameworks where specialized agents query disjoint memory segments can reconcile conflicting information and resolve ambiguities before context injection. Future research must therefore focus on unified retrieval objectives that jointly optimize for semantic relevance, temporal coherence, and computational efficiency, ensuring that external memory serves as a precise extension of the agent's cognitive horizon rather than a source of distraction. While these mechanisms determine *what* information is accessed and *how* it is injected, they set the stage for the subsequent discussion on the dynamic read-write operations that govern the lifecycle of this knowledge, transitioning raw traces into consolidated representations over time.

4.3 Dynamic Read-Write Operations and Memory Consolidation

Dynamic read-write operations constitute the procedural backbone of external memory systems, governing how agents encode, update, and refine their knowledge base over extended horizons. Unlike static parametric storage, these mechanisms facilitate a continuous lifecycle of information, transitioning raw episodic traces into consolidated semantic representations while maintaining temporal coherence. The foundational challenge lies in the write operation: determining not only what to store but also how to structure it for future retrieval. Frameworks such as [31] address this by extracting knowledge as structured triplets based on Davidsonian semantics, ensuring that memory entries are scalable, aggregatable, and interpretable. Similarly, [30] introduces an explicit read-write module that allows models to dynamically interact with stored knowledge, moving beyond the limitations of implicit parametric memorization to support grounded, factual reasoning.

A critical evolution in this domain is the mechanism of memory consolidation, which mimics biological processes by compressing high-frequency episodic details into abstracted semantic summaries. This transition is essential for preventing memory bloat and enabling efficient long-term retention. [48] implements a bio-inspired updating mechanism grounded in the Ebbinghaus Forgetting Curve, where memories are reinforced or allowed to decay based on recency and significance, effectively filtering noise while preserving salient user personality traits. Furthermore, [38] advances this by organizing token sequences into coherent episodic events using Bayesian surprise and graph-theoretic boundary refinement, allowing for a two-stage retrieval process that bridges similarity-based and temporally contiguous access. Such hierarchical consolidation is echoed in [46], where agents utilize coarse-to-fine attention over chunked memories to "mentally time-travel" without processing intervening distractors, thereby optimizing the signal-to-noise ratio in long-horizon tasks.

Maintaining memory integrity requires robust strategies for selective forgetting and belief updating, particularly in non-stationary environments where prior knowledge may become obsolete or contradictory. Adaptive forgetting policies are crucial; [45] proposes "Expire-Span," a learnable mechanism that identifies and expires irrelevant information, enabling transformers to scale to tens of thousands of timesteps by preserving only critical states. In the realm of belief consistency, [34] augments pre-trained models with a symbolic memory that employs a weighted MaxSAT solver to resolve logical conflicts among beliefs, ensuring that the agent's world model remains coherent despite new, potentially conflicting evidence. This dynamic updating is further refined in [56], which introduces a selective elimination mechanism to remove invalidated information in long-term conversations, significantly enhancing conversational engagingness by preventing the recurrence of outdated facts.

Emerging trends point towards hybrid neuro-symbolic approaches and reinforcement learning-driven management policies. [33] exemplifies the shift toward symbolic precision by utilizing SQL databases as external memory, allowing LLMs to execute complex multi-hop reasoning through generated queries rather than approximate vector retrieval. Conversely, [86] leverages reinforcement learning to optimize memory allocation dynamically, balancing the retention of old versus new class exemplars in incremental learning scenarios. Despite these advances, challenges persist in balancing the trade-off between retention fidelity and computational efficiency. While [41] demonstrates that efficient storage of $O(k)$ bits can yield significant gains over state-of-the-art replay methods, the optimal frequency and granularity of consolidation remain open questions. Future research must therefore focus on unified theoretical frameworks that formally define the bounds of memory capacity and the latency of consolidation operations, ensuring that dynamic read-write systems can support truly autonomous, lifelong agency without succumbing to catastrophic interference or semantic drift.

4.4 Interactive and Hierarchical Memory Navigation

Building upon the dynamic read-write foundations and consolidation policies established previously, interactive and hierarchical memory navigation reframes memory access as an active, multi-step decision process essential for resolving long-horizon tasks. Rather than treating external memory as a passive repository queried via single-shot similarity search, this approach empowers agents to iteratively traverse, refine, and manipulate complex memory structures. This shift is critical when relevant context is distributed across vast temporal spans or requires multi-hop reasoning that exceeds the capacity of dense vector retrieval alone, effectively bridging the gap between static storage and the active agency required for lifelong learning.

A primary manifestation of this paradigm is iterative tree-based traversal, where agents navigate hierarchical summaries to "zoom in" on specific details. [54] introduces MemWalker, a method that constructs a tree of summary nodes from long contexts, allowing the model to iteratively navigate this structure to locate relevant information, thereby enhancing both retrieval precision and explainability. Similarly, [46] proposes Hierarchical Chunk Attention Memory (HCAM), which enables agents to perform coarse-grained attention over chunk summaries before executing detailed attention within selected segments. This mechanism facilitates efficient "mental time-travel," allowing agents to recall specific past events without processing intervening noise, significantly outperforming flat memory architectures in tasks requiring long-term retention.

Complementing hierarchical traversal is the integration of tool-augmented manipulation, where memory access becomes a programmable reasoning step. [33] demonstrates that augmenting LLMs with SQL databases as symbolic memory allows for complex multi-hop reasoning that neural memory mechanisms struggle to support due to error accumulation. By generating SQL instructions, agents can actively filter, join, and aggregate data, transforming memory retrieval into a logical operation. This concept extends to graph-based navigation, as seen in [87], where agents autonomously select tools to traverse knowledge graphs, iteratively updating memory states to resolve complex queries. Furthermore, [88] formulates the action space of tool usage as a decision tree, employing A* search to efficiently prune high-cost branches and identify optimal reasoning paths, thus balancing exploration and exploitation in expansive memory-action spaces.

These interactive strategies often synergize with planning frameworks to mitigate the limitations of fixed context windows, effectively operationalizing the neuro-symbolic and RL-driven management policies discussed earlier. [32] conceptualizes this through virtual context management, inspired by operating system hierarchies, where the agent actively manages data movement between fast (context window) and slow (external storage) memory tiers using interrupts and function calls. In embodied domains, [89] utilizes a milestone-based tracker to guide agents through long instruction sequences, effectively navigating the memory of completed sub-tasks to determine progress. Additionally, [90] induces reusable workflows from past experiences, selectively providing these structured memory traces to guide future action generation, thereby reducing the search space for long-horizon web navigation.

Despite these advancements, challenges remain in optimizing the trade-off between the computational cost of iterative reasoning and the fidelity of retrieved information. Current approaches often rely on heuristic stopping criteria or fixed depth limits, which may prematurely terminate deep reasoning chains. Future research must therefore focus on developing adaptive navigation policies that dynamically adjust the granularity of memory access based on task uncertainty and information density, potentially leveraging meta-cognitive signals to determine when to "dig deeper" versus when to proceed with available context. Addressing these optimization hurdles is prerequisite to deploying agents in open-ended environments, setting the stage for the subsequent discussion on evaluation benchmarks and real-world application scenarios.

5 Dynamic Memory Management and Continual Adaptation

5.1 Context Compression and Latent Summarization Strategies

Context compression and latent summarization strategies constitute a critical frontier in enabling long-horizon agency, addressing the fundamental bottleneck where linearly growing interaction histories exceed finite context windows. The primary objective is to distill extensive temporal sequences into compact, information-dense representations that preserve semantic fidelity and procedural knowledge while drastically reducing token counts. A dominant paradigm involves learnable summary vectors and soft prompts, which substitute raw text with continuous latent embeddings. The In-context Autoencoder (ICAE) exemplifies this approach, leveraging a lightweight encoder to compress long contexts into a fixed number of memory slots that are directly conditioned upon by the decoder, achieving significant compression rates with minimal parameter overhead [50]. Similarly, the Memory of Amortized Contexts (MAC) framework compresses unseen documents into compact modulations stored in a memory bank, allowing frozen models to adapt online via amortized feature extraction without gradient updates [91]. These methods effectively transform the memory management problem from discrete token retention to continuous state optimization, though they risk losing fine-grained syntactic details inherent in the original text.

Complementary to latent vector compression are dynamic filtering mechanisms that optimize the signal-to-noise ratio within the active context. Query-aware sparsity approaches, such as Quest, operate on the premise that token criticality is contingent upon the current query, selectively loading only the top-k relevant Key-Value (KV) cache pages to accelerate inference while maintaining accuracy [52]. This aligns with findings that attention matrices in long-context scenarios exhibit inherent sparse patterns (e.g., A-shape, Vertical-Slash), which can be exploited to discard redundant information during the pre-filling stage [7]. Furthermore, prompt compression techniques like LongLLMLingua actively prune irrelevant tokens based on perplexity and semantic density prior to model ingestion, demonstrating that removing up to 90% of context tokens can paradoxically improve performance by mitigating the "lost-in-the-middle" phenomenon and enhancing the model's focus on key information [71]. Such strategies suggest that raw context length is less critical than the density of relevant information, challenging the necessity of processing full sequences for effective reasoning.

Hierarchical chunking and abstracted memory architectures offer a structural solution by organizing memory into multi-resolution layers, facilitating efficient "mental time travel." The MemWalker approach constructs a tree of summary nodes, allowing agents to iteratively navigate from coarse-grained event summaries to detailed episodic traces, thereby bypassing the need to attend to the entire history simultaneously [54]. This mirrors human-like episodic memory systems, such as EM-LLM, which segment token streams into coherent events using Bayesian surprise and graph-theoretic boundary refinement, enabling retrieval based on both semantic similarity and temporal contiguity [38]. Additionally, hierarchical context merging schemes like HOMER employ divide-and-conquer algorithms to progressively merge adjacent chunks at deeper transformer layers, optimizing memory usage logarithmically with respect to input length [73].

Despite these advances, significant trade-offs remain between compression fidelity and computational efficiency. While latent methods offer constant memory footprints, they often require auxiliary training and may suffer from information bottlenecks that obscure rare but critical details. Conversely, pruning and hierarchical methods retain more interpretability but depend heavily on the quality of the segmentation or relevance scoring functions. Future research must converge on hybrid architectures that dynamically switch between lossless retrieval for factual precision and lossy latent summarization for strategic planning, ensuring robustness across varying temporal scales and task complexities.

5.2 Adaptive Forgetting Policies and Salience-Based Pruning

Building upon the compression and structural organization strategies discussed previously, effective long-horizon agency necessitates a complementary mechanism: the strategic discarding of information to prevent cognitive overload and maintain adaptability in non-stationary environments. While prior sections addressed how to distill and structure extensive histories, adaptive forgetting policies specifically manage the lifecycle of these stored memories, resolving the tension between retaining high-utility information and evicting obsolete data. A primary paradigm in this domain involves bio-inspired decay models, where memory retention is governed by temporal and access-based metrics. For instance, [48] implements a mechanism grounded in the Ebbinghaus Forgetting Curve, allowing agents to reinforce significant memories while naturally decaying those that are infrequently accessed or temporally distant, thereby mimicking human-like selectivity. Similarly, [55] demonstrates that in perpetual learning settings, statistical recall biases can drive a “use it or lose it” process, where frequently recalled patterns are preserved while rare ones are forgotten, optimizing the memory footprint for current task demands.

Beyond simple recency-frequency heuristics, advanced frameworks leverage uncertainty and salience to guide more nuanced pruning decisions, moving from static rules to dynamic, learned behaviors. [45] introduces Expire-Span, a learnable mechanism that predicts the importance of past states, enabling Transformers to expire irrelevant information and efficiently attend to tens of thousands of timesteps. This approach contrasts with static eviction policies by learning to retain critical information required for long-range dependencies while discarding redundant context. In the realm of key-value cache management for inference, [92] proposes a general framework combining proxy token eviction with diversified random strategies to mitigate attention bias, significantly reducing memory consumption while preserving performance on long-context tasks. Furthermore, [93] offers an alternative to hard eviction by adaptively merging similar KV states, thus compressing the memory representation without outright deletion, which is particularly effective under constrained memory budgets.

Salience-based pruning also plays a critical role in resolving knowledge conflicts and maintaining logical consistency, a prerequisite for the systemic stability discussed in subsequent sections. [56] highlights the necessity of selectively eliminating invalidated or outdated information in multi-session dialogues to prevent conversational breakdowns, showing that dynamic updates outperform static storage. This is complemented by [34], which employs a reasoning component to revise conflicting beliefs within a symbolic memory, ensuring that the agent's stored knowledge remains logically coherent over time. Moreover, [37] incorporates mechanisms for selective fact forgetting to facilitate one-shot knowledge updates, demonstrating that controlled forgetting is essential for accurate model editing and adaptation.

However, current approaches face challenges in balancing aggressive pruning with the preservation of latent, long-term dependencies. While methods like [46] enable efficient retrieval via hierarchical chunking, they rely on robust underlying pruning policies to prevent the dilution of high-level summaries with noise. Future research must therefore focus on developing context-aware forgetting policies that can distinguish between temporarily irrelevant information and permanently obsolete data, potentially integrating meta-learning strategies to adapt forgetting rates to specific environmental dynamics. The synthesis of uncertainty-driven eviction, salience-based merging, and belief-consistent updating represents a promising trajectory for building resilient, lifelong learning agents. These mechanisms provide the essential groundwork for addressing the broader challenge of catastrophic forgetting, where the goal shifts from managing individual memory items to preserving global competencies across continuous learning streams.

5.3 Mitigation of Catastrophic Forgetting in Continual Adaptation

The mitigation of catastrophic forgetting represents the central challenge in enabling long-horizon LLM agents to operate continuously without degrading previously acquired competencies. This stability-plasticity dilemma is particularly acute in non-stationary environments where agents must integrate novel experiences while retaining foundational knowledge, often without access to original training distributions. Contemporary approaches generally bifurcate into regularization-based parameter isolation, generative replay mechanisms, and architectural modulations that decouple fast and slow learning dynamics.

Regularization techniques aim to constrain parameter updates to subspaces orthogonal to those critical for prior tasks. The Instruction Vector (IV) framework posits that forgetting stems not from erasure but from the overlay of specialized reasoning patterns, proposing IV-guided training to preserve the original computation graph [94]. Similarly, magnitude-based gradient updating (MIGU) leverages the inherent L1-normalized magnitude distributions in linear layers to selectively update parameters, effectively unlocking innate continual learning abilities without requiring task labels or rehearsal data [95]. In reinforcement learning contexts, modulating masks offer a parameter-efficient alternative by learning task-specific masks over a fixed backbone, allowing for the linear combination of previous knowledge to accelerate new task acquisition [96]. These methods excel in parameter efficiency but often struggle when task boundaries are or when the capacity of the "important" parameter subspace is exhausted.

When original data is inaccessible, generative replay has emerged as a potent strategy for synthesizing pseudo-experiences. Self-Synthesized Rehearsal (SSR) utilizes the base LLM to generate synthetic instances via in-context learning, which are then refined by the updated model to create a diverse rehearsal buffer, effectively preserving generalization capabilities [97]. Extending this to latent spaces, the Lifelong VAEGAN (L-VAEGAN) framework integrates generative replay with variational inference to learn meaningful latent representations across domains, enabling interpolation and inference that traditional replay methods cannot support [98]. In embodied settings, model-free generative replay has demonstrated that "hidden replay" architectures can prevent drift in feature-to-action mappings, achieving near-expert performance in complex domains like

StarCraft-2 with minimal sample requirements [99]. However, the fidelity of synthesized data remains a bottleneck; poor generative quality can lead to "hallucinated" replay that reinforces errors rather than stabilizing knowledge.

Architectural solutions increasingly favor dual-memory or modular systems that physically or functionally isolate new learning. The WISE framework addresses the "impossible triangle" of reliability, generalization, and locality in model editing by employing a dual parametric memory scheme—a main memory for pretrained knowledge and a side memory for edits—orchestrated by a learned router [29]. This mirrors biological complementary learning systems, where rapid episodic encoding occurs alongside slow semantic consolidation. Similarly, Larimar introduces a distributed episodic memory allowing for dynamic, one-shot knowledge updates without retraining, effectively decoupling memory storage from parameter weights [37]. For agents operating in open-ended task spaces, the CLIN architecture utilizes a persistent textual memory centered on causal abstractions, enabling continual improvement across varied environments without parameter updates [100].

Despite these advances, significant challenges persist. Most methods assume discrete task boundaries, whereas real-world agent interaction involves continuous, unsegmented data streams. Future research must prioritize task-free adaptation mechanisms, such as confidence-aware moving averages that update representations without explicit boundary signals [101], and explore hybrid neuro-symbolic approaches that combine the fluidity of vector memories with the structural rigidity of symbolic graphs to ensure long-term consistency.

5.4 Multi-Timescale Memory Architectures for Non-Stationary Dynamics

Building upon the architectural foundations that decouple memory storage from parameter weights, multi-timescale memory architectures provide the dynamic mechanism necessary for operating in non-stationary environments where data distributions and agent dynamics shift unpredictably. While previous approaches focused on isolating knowledge to prevent interference, these systems explicitly decouple retention mechanisms across varying temporal horizons to simultaneously resolve immediate contextual fluctuations and preserve stable long-term world models. A foundational strategy involves cascaded replay buffers that accumulate experiences at distinct decay rates, enabling agents to reconcile short-term adaptation needs with long-term knowledge consolidation without relying on explicit task boundary signals [41]. This hierarchical structuring mirrors biological systems where fast-adapting episodic traces are gradually compressed into slow-consolidating semantic representations, thereby mitigating catastrophic forgetting while maintaining responsiveness to novel stimuli [102].

To address the specific challenges of piecewise stable contexts common in real-world robotics and control, recent frameworks have evolved to employ belief-augmented state modeling. For instance, the SeCBAD method jointly infers latent context beliefs and segment lengths, enabling policies to adapt to abrupt environmental variations by dynamically weighting historical data based on its relevance to the current regime [103]. Similarly, in multi-agent settings where opponent strategies induce non-stationarity, multi-timescale learning algorithms permit concurrent policy updates at different rates; faster timescales capture transient strategic shifts, while slower timescales preserve robust cooperative baselines, effectively stabilizing convergence in decentralized environments [104].

Theoretical advancements further suggest that optimal prediction in scale-free signals necessitates "fuzzy" memory systems that sacrifice fine-grained temporal accuracy to represent exponentially long timescales within finite resources [49]. This principle is operationalized in architectures like HiPPO, which projects continuous signals onto polynomial bases to maintain optimal online compressions of history, theoretically avoiding priors on specific timescales and ensuring robustness to out-of-distribution temporal shifts [64]. Complementing this, dynamic context-aware models utilize differentiable queue mechanisms to explicitly memorize correlations between long trajectories, integrating static scene layouts with dynamic temporal coherence to improve motion prediction in complex, interacting agent scenarios [105]. Furthermore, Bayesian non-parametric frameworks, such as those utilizing Dirichlet process mixtures, allow agents to instantiate new environment models dynamically as distributions shift, retrieving old models when previously seen contexts reoccur, thus facilitating scalable lifelong adaptation [106; 107].

Despite these advances, critical challenges remain in balancing the computational overhead of maintaining multiple memory scales against the fidelity of retrieved information. Current approaches often struggle to automatically determine the optimal number of timescales or the precise decay functions required for specific domains without manual tuning. Future research must pivot towards self-regulating architectures that can meta-learn timescale parameters and dynamically allocate memory resources based on the observed entropy of the environment, moving beyond fixed hierarchical structures to fully adaptive, fluid memory systems capable of navigating open-ended, non-stationary task spaces [108].

6 Memory-Enhanced Planning, Reasoning, and Self-Reflection

6.1 Hierarchical Planning and Workflow-Guided Decomposition

Hierarchical planning serves as a critical mechanism for bridging the gap between abstract, long-horizon goals and concrete, low-level actions, particularly when agents must operate within constrained context windows. By decomposing complex objectives into manageable sub-tasks, memory-enhanced agents can maintain strategic coherence while mitigating the cognitive load associated with extended temporal dependencies. A primary paradigm in this domain involves the construction of graph-based representations to model task dependencies and state transitions dynamically. For instance, [109] proposes learning sparse, multi-step transitions as graph edges between latent landmarks, enabling robust planning over high-dimensional continuous control tasks where step-by-step world models typically diverge. Similarly, [110] introduces a data structure that aggregates states based on two-way consistency, theoretically proving that such merging strategies scale linearly with merging thresholds while significantly outperforming baseline methods in sparse-reward visual navigation. These graph-structured approaches allow agents to "zoom out" from immediate sensory inputs to reason over topological relationships, effectively compressing long histories into navigable semantic maps.

Complementing structural representations, workflow-guided decomposition leverages historical experience to induce reusable task routines. The [90] framework exemplifies this by inducing commonly reused workflows from past trajectories and selectively injecting them into the agent's context to guide future generations. This approach substantially improves success rates in web navigation benchmarks by reducing the search space of valid action sequences, demonstrating that explicit memory of procedural knowledge is superior to retrieving raw episodic traces. Furthermore, hierarchical memory management systems like [20] utilize subgoals as memory chunks, allowing agents to proactively replace obsolete observations with summarized states relevant to the current subgoal. This dynamic pruning ensures that the working memory remains focused on immediate strategic intent, achieving a twofold increase in success rates for long-horizon tasks compared to flat history retention.

The integration of formal domain knowledge further refines this decomposition process. Methods such as [111] and [112] highlight the efficacy of translating natural language goals into structured planning languages (e.g., PDDL), where classical solvers handle the low-level sequencing while LLMs manage high-level intent and constraint specification. This neuro-symbolic synergy addresses the hallucination tendencies of pure LLM planners by grounding decomposition in verifiable logical constraints. Additionally, [113] employs scene graphs to ground environmental topology, decomposing long-term goals into autoregressive sub-goals that an automated planner can solve efficiently. Despite these advances, challenges remain in balancing the granularity of decomposition with the fidelity of world models; overly coarse abstractions may miss critical physical constraints, while fine-grained tracing risks re-introducing context bottlenecks. Future research must therefore focus on adaptive hierarchical mechanisms that dynamically adjust the depth of decomposition based on task complexity and environmental uncertainty, potentially leveraging latent space predictors [114] to simulate sub-goal feasibility before commitment.

6.2 Retrospective Reasoning and Iterative Self-Reflection Loops

Building upon the adaptive hierarchical structures and scaffolding established in the previous section, retrospective reasoning and iterative self-reflection constitute the critical cognitive loop that transforms raw interaction histories into actionable policy improvements. While hierarchical planning provides the structural backbone for decomposing long-horizon goals, self-reflection mechanisms enable agents to dynamically retrieve past trajectories, diagnose deviations from intended sub-goals, and synthesize corrected strategies for future execution. The core challenge in this domain lies in effectively leveraging both successful and failed experiences to refine decision-making without incurring prohibitive computational costs or suffering from catastrophic forgetting.

A primary approach involves trajectory-based error analysis, where agents explicitly reconstruct the causal chain of failure to update their internal policies. [115] demonstrates that incorporating unsuccessful trajectories during fine-tuning, marked by specific indicators of failure, significantly enhances agent robustness compared to training solely on expert demonstrations. This aligns with the principles of [116], which proposes an iterative framework where agents learn from contrastive trajectory pairs generated during exploration, thereby optimizing policies through direct exposure to failure modes. Furthermore, [117] introduces a principled method where a retrospective model summarizes the root causes of prior failures to refine prompts via policy gradient, effectively translating environment-specific rewards into linguistic feedback loops.

Complementing these post-hoc analyses, anticipatory reflection mechanisms simulate potential outcomes before action execution to preempt errors, creating a bridge toward the predictive world modeling discussed in the subsequent section. [118] integrates future prediction with reasoning and acting, allowing agents to diversify their strategic thinking by evaluating the divergence between predicted and actual states. This “reason for future, act for now” paradigm is formalized in [119], which casts reasoning as Bayesian adaptive planning, achieving provable regret bounds by continuously updating posteriors based on memory buffers. Similarly, [120] leverages closed-loop language feedback to form an internal monologue, enabling robots to richer process environmental changes and adjust plans in real-time.

The efficacy of these loops often depends on the depth and granularity of the reflection process. [121] highlights the necessity of context-aware, iterative prompting to elicit multi-step reasoning capabilities that static prompts cannot achieve. At a higher level of abstraction, [122] moves beyond action-level correction to reflect on entire belief systems and policies, employing depth-first search to optimize behavioral strategies in complex dynamic environments like poker. This mirrors the recursive introspection proposed in [123], which fine-tunes models to recursively detect and correct mistakes over multiple turns, effectively treating self-improvement as a multi-turn Markov Decision Process.

However, significant trade-offs remain regarding the stability and scalability of self-reflection. While [124] confirms that self-reflection generally improves accuracy, the quality of reflection is highly sensitive to the prompting strategy and the model's inherent reasoning capacity. Moreover, unchecked feedback loops can lead to in-context reward hacking, where agents optimize for superficial metrics at the cost of genuine task completion, as warned in [125]. Future research must therefore focus on developing verifiable reflection metrics and hybrid neuro-symbolic critics, such as those in [126], to ensure that iterative self-reflection leads to genuine cognitive advancement rather than overfitting to spurious environmental cues. Addressing these limitations is essential for transitioning from reactive correction to the proactive, simulation-based agency explored in the following section, where memory serves not just as a record of the past, but as a generative engine for imagining futures.

6.3 World Modeling and Latent State Prediction via Memory

World modeling via memory empowers agents to construct internal simulations of environment dynamics, enabling long-horizon prediction and planning in partially observable settings where raw context windows fail. By decoupling the learning of world dynamics from policy execution, agents can leverage stored episodic and semantic memories to anticipate future states, evaluate counterfactuals, and refine strategies without costly real-world interactions. This paradigm shifts the role of memory from passive retrieval to active generative simulation, where latent representations encapsulate complex temporal dependencies essential for robust agency.

A primary approach involves learning compact latent space models that summarize high-dimensional sensory inputs into manageable state representations. Frameworks like Dreamer demonstrate that agents can solve long-horizon tasks purely through "latent imagination," propagating analytic gradients through trajectories simulated within a learned world model to optimize behavior efficiently [127]. To address the divergence of predictions over extended horizons, recent advances integrate recurrent structures with explicit memory mechanisms. For instance, Recall to Imagine (R2I) incorporates state space models (SSMs) into world models to enhance long-term credit assignment and temporal coherence, achieving superhuman performance in memory-intensive domains like Memory Maze [128]. Similarly, transformer-based world models (TWM) utilize attention mechanisms to access previous states directly rather than through compressed recurrent hidden states, allowing for flexible reasoning over rewards and actions across 100k interactions [129]. These architectures effectively mitigate the "sharp nearby, fuzzy far away" phenomenon by maintaining distinct representations for immediate and distant temporal contexts.

Beyond passive prediction, advanced agents employ memory to facilitate counterfactual reasoning and "what-if" simulations. The Variational Temporal Abstraction (VTA) framework enables jumpy imagination by inferring latent temporal structures, allowing agents to skip irrelevant steps and plan more efficiently in hierarchical state spaces [130]. This capability is further extended in systems like MineDreamer, which utilizes a Chain-of-Imagination mechanism to envision step-by-step execution processes, translating abstract instructions into precise visual prompts and control signals [131]. Such imaginative capacities are critical for handling uncertainty; DESIRE, for example, employs a conditional variational autoencoder to generate diverse hypothetical future samples, which are then ranked based on accumulated rewards to make strategic predictions in dynamic scenes [132]. By storing these diverse trajectories in memory, agents can evaluate multiple potential outcomes before committing to an action, significantly reducing the risk of catastrophic failures in complex environments.

However, reliance on learned world models introduces challenges regarding model bias and hallucination. Purely neural simulators may drift from reality when rolled out over long horizons, necessitating hybrid approaches that ground simulations in explicit knowledge. The Reasoning via Planning (RAP) framework repurposes LLMs as both world models and reasoning agents, using Monte Carlo Tree Search to explore reasoning paths guided by task-specific rewards, thereby balancing exploration and exploitation in vast state spaces [133]. Furthermore, integrating symbolic structures can enhance fidelity; L3P learns graph-structured world models composed of sparse, multi-step transitions between latent landmarks, enabling temporally extended reasoning that outperforms step-by-step rollout methods in continuous control tasks [109]. Similarly, optimizing for long-term future consistency through auxiliary tasks forces latent variables to carry forward-looking information, improving the reliability of long-horizon predictions [134].

Despite these advances, a fundamental trade-off remains between the flexibility of neural world models and the precision of symbolic planners. While neural models excel at handling noisy, high-dimensional observations, they often lack the rigorous consistency required for verifiable planning. Future research must therefore focus on neuro-symbolic integration, where memory-augmented neural simulators provide heuristic guidance to formal planners, or where explicit memory graphs constrain the generative space of LLM-based world models. Additionally, developing metrics to quantify the "dream-reality gap" in latent simulations will be crucial for ensuring that agents can distinguish between plausible imaginings and physically grounded predictions, ultimately leading to more trustworthy autonomous systems.

6.4 Memory-Augmented Search and Strategic Exploration

Bridging the gap between the neural world models discussed previously and the metacognitive architectures to follow, the integration of memory mechanisms with algorithmic search represents a pivotal advancement for LLM agents navigating complex, long-horizon decision spaces. While purely neural simulators excel at handling noisy observations, they often lack the rigorous consistency required for verifiable planning; conversely, standard autoregressive generation frequently succumbs to local optima due to its greedy nature. Memory-augmented search frameworks resolve this tension by leveraging historical trajectories to guide strategic exploration, effectively balancing the exploitation of known successful paths with the exploration of novel reasoning branches. A primary paradigm in this domain involves adapting Monte Carlo Tree Search (MCTS) to the linguistic realm, where memory serves as both a heuristic prior and a repository for value estimation. For instance, the Reasoning via Planning (RAP) framework repurposes the LLM as a world model to simulate future states within a tree structure, utilizing retrieved high-reward trajectories to bias node expansion and accelerate convergence in sparse-reward environments [133]. Similarly, AlphaLLM integrates MCTS with self-critique loops, where memory buffers store successful reasoning paths that are subsequently used to refine the policy without external annotations, demonstrating that strategic search can induce self-improvement [135].

Beyond tree-based methods, memory facilitates efficient navigation through expansive action spaces by pruning irrelevant branches based on historical consistency. The ToolChain* approach formulates tool usage as a decision tree, employing A* search guided by cost functions derived from past execution failures to identify optimal API call sequences, thereby mitigating the combinatorial explosion inherent in multi-step tool interaction [88]. In reinforcement learning contexts, agents leverage episodic memory to generalize across continuous state spaces where exact state revisitation is improbable. Generalizable Episodic Memory (GEM) organizes state-action values to support implicit planning over memorized trajectories, significantly enhancing sample efficiency in continuous control tasks by allowing agents to "plan" over past experiences rather than relying solely on parametric knowledge [136]. Furthermore, the synergy between memory and search extends to neuro-symbolic architectures, where external symbolic planners utilize LLM-generated memory to initialize search states or provide heuristic guidance. The LLM+P framework exemplifies this by translating natural language problems into formal planning domains, where classical solvers exploit the structured memory of the problem state to guarantee optimality, a capability often absent in pure LLM approaches [111].

Despite these advances, significant challenges remain regarding the scalability and fidelity of memory-informed search, echoing the latency concerns that motivate the shift toward adaptive metacognition in the subsequent section. Current methods often struggle with the "curse of history," where the retrieval of irrelevant or outdated memories can misguide the search process, leading to suboptimal planning in dynamic environments [18]. Moreover, the computational overhead of maintaining and querying large memory banks during iterative search steps poses a latency bottleneck for real-time applications. Emerging trends suggest a shift towards hierarchical memory structures that allow for coarse-to-fine search strategies, akin to the "mental time travel" observed in biological systems, where agents first retrieve abstracted summaries to guide high-level planning before drilling down into detailed episodic traces for execution [46]. Future research must address the development of adaptive memory pruning policies that dynamically adjust the granularity of stored information based on search depth and uncertainty. This evolution from static retrieval to dynamic, context-aware memory management lays the groundwork for the higher-order self-regulation and strategy modulation mechanisms explored next, ensuring that strategic exploration remains computationally tractable while preserving the critical long-range dependencies necessary for robust long-horizon agency.

6.5 Metacognition and Adaptive Strategy Modulation

Metacognition and adaptive strategy modulation represent the apex of cognitive architecture in long-horizon agents, transcending static memory retrieval to enable dynamic self-regulation of reasoning processes. This domain focuses on higher-order mechanisms where agents continuously monitor their internal epistemic states and memory utilization efficiency to modulate planning strategies, allocate computational resources, and adapt to non-stationary environmental dynamics. Unlike conventional agents that execute fixed prompt chains, metacognitive agents instantiate a feedback loop wherein the agent evaluates the confidence and consistency of its ongoing plans, triggering strategy revision protocols when uncertainty exceeds predefined thresholds.

A fundamental component of this architecture is the maintenance of an internal working memory buffer that facilitates rapid task switching while preventing interference between concurrent goals. Inspired by biological systems where evolution discovers mechanisms for integrating sensory signals with ongoing internal activity to support alternative actions [137], artificial agents utilize hybrid memory dynamics to manage multi-timescale dependencies. For instance, frameworks employing both leaky and non-leaky units allow agents to discard transient distractors while preserving high-level strategic context, effectively solving hierarchical tasks that require distinct retention scales [27]. This selective retention is further refined by "use it or lose it" policies, where statistical recall biases during perpetual learning ensure that frequently accessed strategies are reinforced while obsolete memories decay, mimicking the stimulus-driven forgetting observed in biological systems [55].

Adaptive strategy modulation is increasingly realized through recursive introspection and self-improvement loops. Approaches such as RISE (Recursive IntroSpEction) formalize self-correction as a multi-turn Markov Decision Process, enabling agents to iteratively detect and rectify reasoning errors based on environmental feedback [123]. Similarly, the concept of "Memory-of-Thought" allows agents to pre-compute and store high-confidence reasoning traces as external memory, which are subsequently retrieved to augment test-time reasoning without parameter updates [138]. These mechanisms are critical for mitigating the "chain of thoughtlessness" phenomenon, where standard prompting fails to generalize out-of-distribution due to a lack of algorithmic procedure learning [139]. By integrating retrieval-augmented thoughts, agents can revise their reasoning steps iteratively, significantly reducing hallucination in long-horizon generation tasks [84].

Furthermore, advanced agents employ policy-level reflection to evolve their behavioral strategies over time. Agent-Pro, for example, utilizes depth-first search combined with belief reflection to optimize policies dynamically, demonstrating superiority in complex interactive games where static prompts fail [122]. This aligns with the "reason for future, act for now" paradigm, where agents maintain a memory buffer to update their posterior beliefs about the environment, achieving provable regret bounds in Bayesian adaptive MDPs [119]. The integration of layered memory structures, as seen in financial trading agents like FinMem, further illustrates how adjustable cognitive spans enable agents to prioritize critical information and self-evolve professional knowledge in volatile environments [140].

Despite these advances, challenges remain in balancing the computational cost of metacognitive monitoring with inference latency. Future research must explore unified theoretical frameworks that treat metacognition not as an add-on module but as an emergent property of meta-reinforcement learning, where transformers inherently function as Bayes-optimal agents [141; 142]. Developing standardized metrics for quantifying the "metacognitive gap"—the disparity between an agent's perceived and actual competence—will be essential for advancing robust, self-adaptive long-horizon agency.

7 Evaluation Frameworks, Benchmarks, and Empirical Analysis

7.1 Synthetic and Naturalistic Benchmarks for Long-Range Dependency

The evaluation of long-range dependency mechanisms in large language model (LLM) agents necessitates a rigorous dichotomy between synthetic stress tests, which isolate specific retrieval capabilities, and naturalistic benchmarks, which assess holistic reasoning within complex, domain-specific contexts. Synthetic benchmarks serve as the foundational stress tests for context windows, primarily utilizing the "needle-in-a-haystack" (NIAH) paradigm to measure an agent's ability to locate specific information amidst varying degrees of distractor noise. While early iterations of NIAH provided basic retrieval metrics, recent advancements have introduced significantly more demanding configurations. For instance, [143] extends evaluation beyond 100k tokens, revealing that existing models struggle to effectively process such extreme lengths despite claims of large context windows. Similarly, [144] expands upon vanilla NIAH by incorporating multi-hop tracing and aggregation tasks, demonstrating that while models may achieve near-perfect accuracy on single-fact retrieval, their performance degrades precipitously as task complexity and sequence length increase. This limitation is further corroborated by [145], which introduces the Ancestral Trace Challenge to mimic real-world logical reasoning, highlighting a critical gap between simple retrieval and deep contextual synthesis. Furthermore, [67] pushes these boundaries to 11 million tokens, showing that recurrent memory augmentations can outperform standard transformer-based approaches in extracting distributed facts from extensive texts, thereby establishing a new ceiling for sequence processing capabilities.

In contrast to synthetic isolation, naturalistic benchmarks evaluate long-context understanding within coherent, high-stakes domains such as literature, law, and interactive dialogue. These datasets require agents to maintain state consistency and perform global reasoning over book-length inputs rather than isolated token retrieval. [146] provides a comprehensive bilingual, multitask framework covering single-document QA, summarization, and code completion, establishing that while scaled position embeddings improve performance, context compression techniques often lag behind native long-context capabilities. The challenge of global reasoning is starkly illustrated in [147], where models fail to verify claims requiring synthesis across entire fictional books, performing no better than random chance compared to human readers. This deficiency extends to conversational agents; [148] evaluates very long-term dialogues spanning dozens of sessions, revealing that current LLMs struggle to comprehend long-range temporal and causal dynamics even when equipped with retrieval-augmented generation. Additionally, [149] addresses the inadequacy of standard n-gram metrics by advocating for length-instruction-enhanced evaluation, noting that popular metrics often fail to correlate with human judgment in long-context scenarios. The domain specificity of long-range utility is further emphasized by [150], which finds that extended context primarily benefits literary novels rather than textbooks, suggesting that the utility of long-context mechanisms is highly dependent on the inherent coherence of the input data.

Emerging trends indicate a shift from passive retrieval assessment to active, multi-step reasoning evaluation. Benchmarks like [151] focus on long in-context learning with extreme label spaces, exposing models' inability to utilize long demonstration sequences for complex classification tasks. Similarly, [41] introduces sequential image reasoning, addressing the multimodal dimension of long-horizon memory where agents must track dynamic object behaviors over time. Despite these advancements, a consistent finding across [44] and [43] is the "lost-in-the-middle" phenomenon, where models exhibit positional bias, favoring information at the beginning or end of sequences while neglecting central content. Future evaluation frameworks must therefore integrate dynamic, interactive environments that test not only static retention but also the agent's ability to update, prune, and reason over evolving memory states, moving beyond static document QA toward embodied, lifelong learning scenarios.

7.2 Embodied and Interactive Evaluation Environments

Building on the identified limitations of static document QA and the necessity for dynamic, interactive environments, embodied and interactive evaluation contexts constitute a critical frontier for assessing memory mechanisms in long-horizon agents. These environments move beyond passive text retrieval to grounded interactions where agents must maintain spatial maps, track dynamic object states, and navigate partially observable worlds. Unlike the synthetic stress tests discussed previously, these settings demand the tight integration of perception, action, and persistent memory to achieve cumulative rewards over extended temporal scales. The landscape of such evaluations is diverse, ranging from controlled 3D navigation tasks to open-ended sandbox simulations that test the limits of an agent's ability to consolidate episodic experiences into semantic knowledge, thereby bridging the gap toward the quantitative efficiency metrics explored in subsequent sections.

A primary category of evaluation focuses on spatial navigation and map construction within partially observable 3D environments. Benchmarks such as [51] and [80] rigorously test an agent's capacity to build and query internal spatial representations over long time lags, often requiring the disentanglement of transient visual inputs from stable environmental layouts. More recent frameworks like [152] introduce dynamic stressors—such as fires or floods—to evaluate how memory systems adapt to non-stationary world states, necessitating rapid belief updates and the pruning of obsolete spatial data. Similarly, [14] and [153] highlight the necessity of attention-based memory architectures that can exploit spatio-temporal dependencies across thousands of steps, demonstrating superior performance over reactive policies in tasks requiring long-range planning.

Beyond pure navigation, open-world sandboxes provide a richer testbed for evaluating hierarchical memory and skill acquisition, pushing the boundaries of context utilization further than static benchmarks. The Minecraft universe has emerged as a dominant platform for this purpose, with frameworks like [154] and [155] assessing agents' abilities to autonomously explore, acquire executable code skills, and manage an ever-growing library of behaviors. These environments expose the limitations of fixed-context windows, necessitating external memory modules that support lifelong learning without catastrophic forgetting. Complementary benchmarks like [156] further stress-test multimodal memory systems, requiring agents to synthesize visual observations and textual instructions into abstracted experience pools for complex task decomposition.

Interactive evaluation also extends to multi-agent and competitive scenarios, where memory is essential for modeling opponent strategies and coordinating team behaviors, adding a layer of social complexity absent in single-agent retrieval tasks. Environments such as [157] and [158] measure an agent's ability to retain historical interaction logs to infer intent and optimize social strategies. Furthermore, [159] provides a multi-dimensional suite including operating system and database interactions, evaluating how agents manage stateful contexts across diverse API calls. Despite these advances, a significant gap remains in standardizing metrics for memory fidelity versus computational cost in real-time embodied settings. Future directions must prioritize unified protocols that decouple memory retrieval latency from action execution speed, ensuring that evaluations reflect the true constraints of deploying long-horizon agents in physical robotics and dynamic virtual worlds, thus setting the stage for the rigorous quantitative assessment of efficiency and utilization detailed next.

7.3 Metrics for Memory Efficiency, Fidelity, and Utilization

Evaluating memory mechanisms in long-horizon agents requires moving beyond aggregate task success rates to quantify the interplay between computational efficiency, retrieval fidelity, and the functional utilization of historical context. A primary dimension of assessment is retrieval precision relative to context scale. While synthetic stress tests like the "needle-in-a-haystack" paradigm provide a baseline for existence proofs, they often fail to capture the degradation of reasoning capabilities as context length increases. Recent benchmarks such as RULER and BABILong demonstrate that models achieving near-perfect retrieval on simple tasks exhibit sharp performance declines when faced with multi-hop tracing or aggregation requirements across extended sequences [6; 68]. Consequently, metrics must evolve to measure "effective context depth," quantifying the maximum sequence length over which an agent can maintain non-trivial reasoning fidelity rather than mere token recall.

Computational efficiency constitutes the second critical pillar, necessitating metrics that balance latency and memory footprint against information preservation. As context windows expand to millions of tokens, the quadratic complexity of attention becomes prohibitive, driving the adoption of sparse attention patterns and KV cache compression techniques. Metrics such as tokens-per-second during pre-filling and peak GPU memory consumption are essential for characterizing systems like

MInference and Quest, which leverage dynamic sparsity to accelerate inference [7; 52]. Furthermore, compression strategies like LongLLMLingua and KVMerger introduce a trade-off between token reduction rates and semantic loss, requiring evaluation protocols that measure the "compression fidelity ratio"—the proportion of task-critical information retained per unit of compressed memory [71; 93]. Without such normalized efficiency metrics, claims of scalability remain ambiguous, particularly when comparing native long-context models against retrieval-augmented architectures where retrieval latency often dominates total response time [160].

Finally, assessing the actual utilization of memory demands diagnostic tools that disentangle rote memorization from active reasoning. The phenomenon of "lost-in-the-middle" suggests that models often fail to integrate information located centrally within long contexts, relying instead on recency biases or parametric priors [36; 43]. To address this, evaluation frameworks must incorporate perturbation analyses and attention visualization to verify that retrieved memories causally influence decision-making steps. For instance, in embodied tasks, metrics should track the correlation between retrieved episodic traces and successful plan execution, ensuring that memory access translates to behavioral adaptation rather than passive storage [161]. Emerging approaches also advocate for budget-aware evaluation, where reasoning strategies are judged not only by accuracy but by their compute efficiency, preventing the conflation of performance gains with brute-force resource allocation [162]. Future research must standardize these multidimensional metrics to rigorously distinguish between architectures that merely store history and those that effectively operationalize it for long-horizon agency.

7.4 Evaluating Believability, Consistency, and Safety in Long-Horizon Agents

Building upon the quantitative foundations of retrieval fidelity and computational efficiency established previously, the evaluation of long-horizon agents must expand to encompass qualitative behavioral attributes, specifically believability, consistency, and safety. While prior metrics quantify how well an agent stores and accesses information, these qualitative dimensions assess whether the operationalization of memory results in coherent, reliable, and aligned agency over extended temporal scales. The accumulation of historical context introduces complex failure modes where persona drift, strategic incoherence, and unsafe trajectory escalation can occur even when individual actions appear benign and retrieval scores remain high. Believability, often quantified through persona stability, requires agents to maintain coherent character profiles and decision-making logics despite perturbations in long input histories. Frameworks such as [163] and [164] have emerged to systematically measure adherence to assigned personas, revealing that increased model scale does not inherently guarantee improved persona fidelity. These studies highlight a critical gap: while large language models possess the capacity for contextual response, they often struggle to sustain distinct behavioral traits over thousands of interaction turns without explicit memory anchoring mechanisms like those proposed in [165] or [41].

Parallel to persona stability, consistency in long-term planning demands that agents align immediate actions with distal goals while adapting to dynamic environmental feedback, ensuring that retrieved memories guide rather than disrupt strategic execution. In high-stakes domains such as financial trading or international event forecasting, evaluated by benchmarks like [166] and [167], consistency is measured by an agent's ability to formulate robust strategies and manage risk based on historical trends rather than reacting myopically to transient noise. The [168] benchmark further elucidates this by simulating competitive auction environments where agents must balance budget management with goal adherence over prolonged sequences. However, empirical evidence suggests that without structured reflection loops or explicit belief state representations, agents frequently succumb to hallucinations regarding task states or become trapped in suboptimal loops, as noted in evaluations of theory of mind capabilities [169]. To mitigate these inconsistencies, frameworks incorporating iterative self-reflection, such as [117] and [170], have demonstrated efficacy in refining plans based on environmental feedback, thereby enhancing the coherence of long-horizon execution.

Finally, safety evaluation in long-horizon contexts extends beyond static content filtering to include the preemptive detection of risky trajectories that emerge from the compounding effects of memory retrieval and sequential decision-making. The [171] framework introduces a constitution-based approach, utilizing pre-planning, in-planning, and post-planning strategies to inject safety constraints directly into the agent's reasoning loop. This is particularly vital in embodied settings where irreversible actions may result from accumulated errors, a risk highlighted in dynamic environments like [152] and [163]. Furthermore, recent work on malfunction amplification attacks [172] underscores the vulnerability of autonomous agents to repetitive or irrelevant action loops induced by adversarial context manipulation, necessitating robust self-examination protocols. The integration of safety metrics with believability and consistency creates a multidimensional evaluation space where an agent must not only perform tasks effectively but also do so in a manner that is predictable, aligned with human values, and resilient to contextual degradation. These behavioral assessments lay the groundwork for the subsequent discussion on human-centric evaluation, where the focus shifts from automated metric validation to fostering transparency, controllability, and trust through interactive memory management.

7.5 Human-Centric and Interactive Memory Management Assessment

The evaluation of memory mechanisms in long-horizon agents has traditionally prioritized automated metrics such as retrieval precision and task success rates. However, as agents transition into collaborative and companionship roles, the adequacy of these metrics diminishes. This subsection argues for a paradigm shift toward human-centric assessment frameworks that evaluate the transparency, controllability, and interpretability of agent memory systems. The core premise is that effective long-term agency requires not only accurate information storage but also a shared mental model between the human user and the artificial agent, enabling seamless interaction and trust.

A primary dimension of this assessment is interactive memory control, which empowers users to inspect and manipulate the agent's internal state. [173] introduces a design probe treating memories as explicit data objects that users can view, edit, summarize, and share, thereby mitigating conversational breakdowns caused by opaque memory management. This approach contrasts with static memory systems where users lack affordances to correct hallucinated or outdated facts, a limitation highlighted in [56], which demonstrates that selective elimination of invalidated information significantly improves perceived humanness and engagingness. Furthermore, the transparency of retrieval paths is critical for verifying logical consistency. [54] proposes methods where agents interactively navigate summary trees to locate information, enhancing explainability by exposing the reasoning steps taken during retrieval. Such interactive reading protocols allow humans to audit whether the agent's focus aligns with their expectations, bridging the gap between black-box attention mechanisms and interpretable cognitive processes.

In collaborative settings, evaluation extends to the efficacy of shared memory states in hybrid intelligence teams. [174] explores frameworks where human intervention is strategically integrated, suggesting that optimal memory management involves dynamic policies for determining when to offload cognitive load to humans versus relying on internal buffers. Similarly, [85] underscores the necessity of robust memory exchange protocols in multi-agent societies, where misaligned memory states can lead to coordination failures. The challenge lies in quantifying the “collaborative gain” attributable to memory sharing; benchmarks like [175] and [176] provide sandboxed environments to measure how effectively agents update shared knowledge graphs or episodic logs to enhance joint task performance compared to solitary operation.

Despite these advances, significant gaps remain in standardizing metrics for subjective qualities such as persona stability and trust. [148] reveals that while models can retrieve facts, they often fail to comprehend long-range temporal and causal dynamics essential for maintaining coherent personas over months of interaction. [164] further argues that increased model scale does not inherently improve persona adherence, necessitating algorithmic innovations in memory updating rather than mere capacity expansion. Additionally, [177] indicates that memory fidelity directly influences trust dynamics in economic games, suggesting that future evaluation frameworks must incorporate behavioral economics metrics to assess how memory errors propagate into strategic distrust. Ultimately, the field must move beyond static benchmarks toward dynamic, longitudinal evaluations where human feedback loops actively shape memory consolidation policies, ensuring agents evolve in alignment with human values and contextual nuances.

8 Conclusion

This subsection synthesizes the architectural and algorithmic trajectories surveyed, positing that robust long-horizon agency emerges not from context extension alone, but from the synergistic integration of native sequence modeling with dynamic, external memory systems. The field has witnessed a dichotomy between approaches that scale the transformer’s attention mechanism to handle massive contexts [42; 53] and those that decouple memory into specialized, retrievable stores [32; 48]. While native extensions like sparse attention patterns [7] and state space models [178] offer linear complexity and infinite theoretical horizons, they often struggle with the “needle-in-a-haystack” retrieval precision required for complex reasoning over millions of tokens [145; 5]. Conversely, retrieval-augmented frameworks provide scalability but introduce latency and potential consistency errors when injecting fragmented context [160]. The most promising trajectory lies in hybrid semi-parametric architectures that treat memory as a hierarchical operating system, dynamically swapping information between fast, limited working memory and slow, vast long-term storage [32; 179].

Despite these advances, critical gaps remain in achieving truly autonomous, lifelong agents. A primary challenge is the “impossible triangle” of reliability, generalization, and locality in memory updates, where editing parametric weights risks catastrophic forgetting while non-parametric retrieval lacks deep semantic integration [29]. Current mechanisms often fail to distinguish between transient episodic traces and consolidated semantic knowledge, leading to memory bloat and degraded performance over extended interactions [56; 148]. Furthermore, existing forgetting policies, though inspired by biological curves [48], lack the adaptive, utility-driven pruning necessary for non-stationary environments, often retaining redundant data while discarding high-uncertainty experiences crucial for learning [40; 180]. The inability to effectively compress long contexts without semantic loss [71; 50] further limits the depth of planning agents can perform, as evidenced by struggles in tasks requiring multi-hop reasoning across 100k+ token windows [143; 151].

Future research must prioritize the development of unified theoretical frameworks that formalize memory operations as continuous belief updates rather than discrete read-write events. We advocate for bio-inspired consolidation mechanisms that actively synthesize episodic experiences into abstract world models, enabling agents to “dream” or rehearse past trajectories to stabilize learning without explicit replay buffers [181; 17]. Additionally, the community requires standardized, lifelong evaluation protocols that move beyond static benchmarks to assess agents in open-ended, evolving environments where task distributions shift over time [182; 183]. Emphasis must also be placed on energy-efficient memory management for edge deployment, leveraging techniques like KV cache compression and distributed attention to make long-horizon agency viable on resource-constrained devices [179; 184]. Ultimately, the path to Artificial General Intelligence depends on transitioning from agents that merely recall the past to those that strategically utilize memory to anticipate, plan, and adapt indefinitely.

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[163] TimeArena Shaping Efficient Multitasking Language Agents in a Time-Aware Simulation

[164] PersonaGym: Evaluating Persona Agents and LLMs

[165] Character-LLM A Trainable Agent for Role-Playing

[166] Understanding LSTM – a tutorial into Long Short-Term Memory Recurrent Neural Networks

[167] Designing Heterogeneous LLM Agents for Financial Sentiment Analysis

[168] Agent Q: Advanced Reasoning and Learning for Autonomous AI Agents

[169] Theory of Mind for Multi-Agent Collaboration via Large Language Models

[170] AdaPlanner Adaptive Planning from Feedback with Language Models

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[172] Breaking Agents: Compromising Autonomous LLM Agents Through Malfunction Amplification

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[174] Large Language Model-based Human-Agent Collaboration for Complex Task Solving

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[177] Can Large Language Model Agents Simulate Human Trust Behaviors

[178] Structured State Space Models for In-Context Reinforcement Learning

[179] LLM as a System Service on Mobile Devices

[180] Improved Schemes for Episodic Memory-based Lifelong Learning

[181] Modular Lifelong Reinforcement Learning via Neural Composition

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[184] Infinite-LLM Efficient LLM Service for Long Context with DistAttention and Distributed KVCache